



Lecture Session 6

1. Water's atoms and molecules and implications to its physical properties;
2. Water's thermal capabilities;
3. Salinity;
4. Density of seawater (Ocean Layers): Temperature, Salinity, and Pressure

Icebreaker

Living, working, and studying near the beach is a privilege of few (especially in San Diego). Some of the reasons this is so pleasant are the constant breeze and the comfortable temperatures .

- Can you explain how the ocean and these amazing conditions are related?



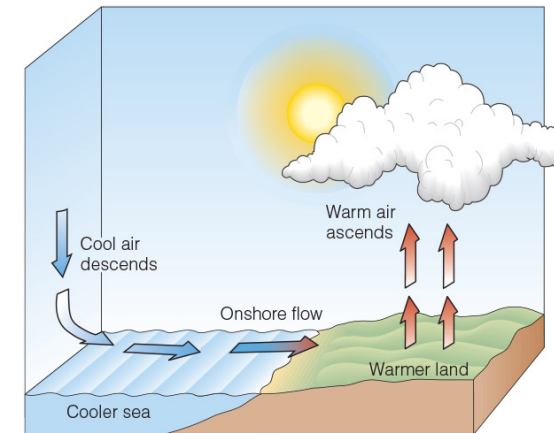
Icebreaker

Living, working, and studying near the beach is a privilege of few (especially in San Diego). Some of the reasons this is so pleasant are the constant breeze and the comfortable temperatures .

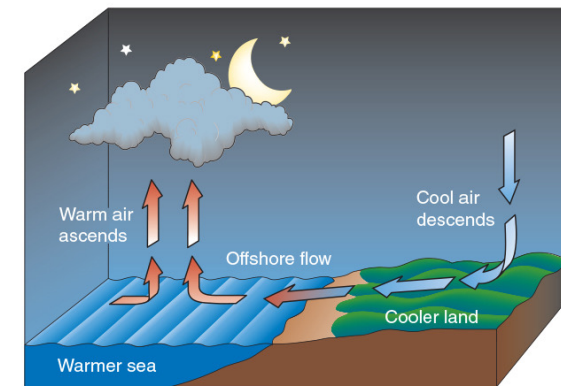
- Can you explain how the ocean and these amazing conditions are related?



The answer is related to water properties: Ex. High Heat Capacity



a In the afternoon, the land is warmer than the ocean surface, and warm air rising from the land is replaced by an onshore sea breeze.



b At night, as the land cools, the air over the ocean is now warmer than the air over the land. The ocean air rises. Air flows offshore to replace it, generating an offshore flow (a land breeze).

Figure 8.20 The flow of air in coastal regions during stable weather conditions.

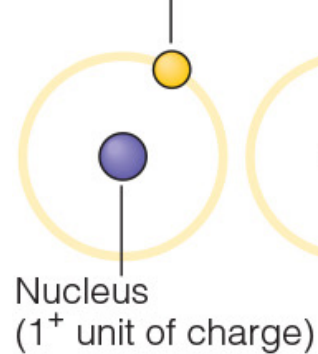
The Water Molecule: H₂O

Protons: + charge
Electrons: - charge

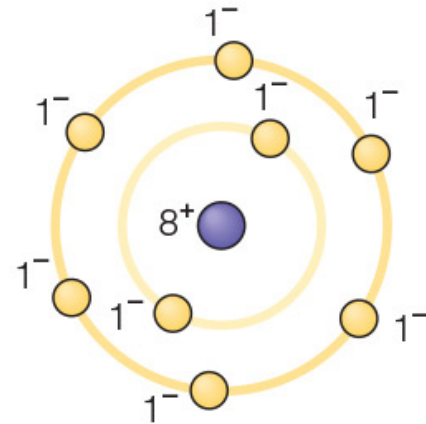
Covalent Bonds = connection
due to sharing of electrons

Two hydrogen atoms . . .

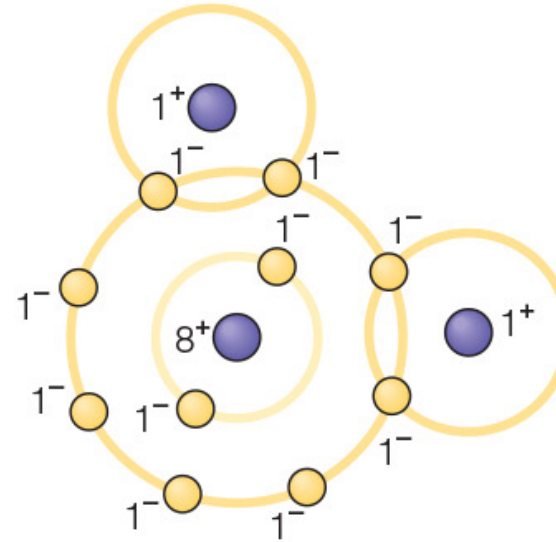
Electron
(1⁻ unit of charge)



share their electrons
with one oxygen atom . . .



to form a water molecule held
together by covalent bonds . . .



which acts as if it has
negative and positive ends.

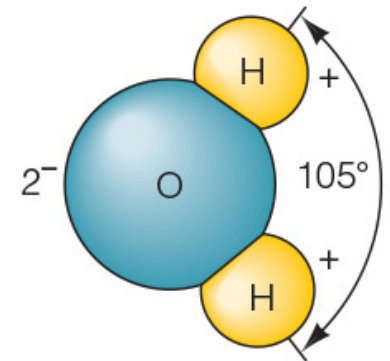


Figure 6.3 The formation of a water molecule. Atoms are most stable when they have a full outer shell of electrons. Hydrogen is most stable when it has two electrons in its outer shell and the larger oxygen atom (like many of the atoms relevant to our discussion of oceanography), is typically most stable with a total of 8 electrons in its outer shell. Since oxygen only has six electrons in its outer shell (because the first two of its eight electrons must fill the inner shell), oxygen bonds to two different hydrogen atoms by sharing hydrogen's electrons so that each atom has a full outer shell.

The Water Molecule: H₂O

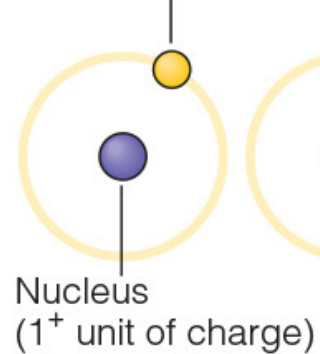
Protons: + charge
Electrons: - charge

Covalent Bonds = connection
due to sharing of electrons

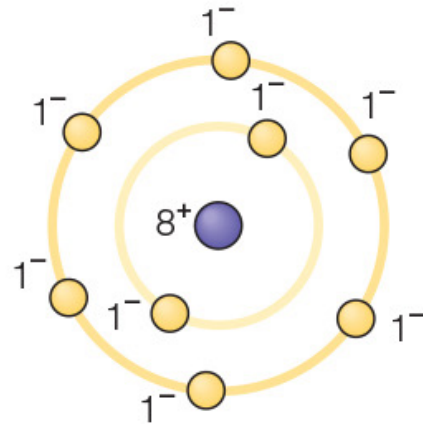
H₂O molecule configuration leaves the H's nucleus (protons) exposed. In contrast, the O's electrons on the other side are also exposed forming a Polar molecule. Therefore, like a magnet, it has a positive and negative charge

Two hydrogen atoms . . .

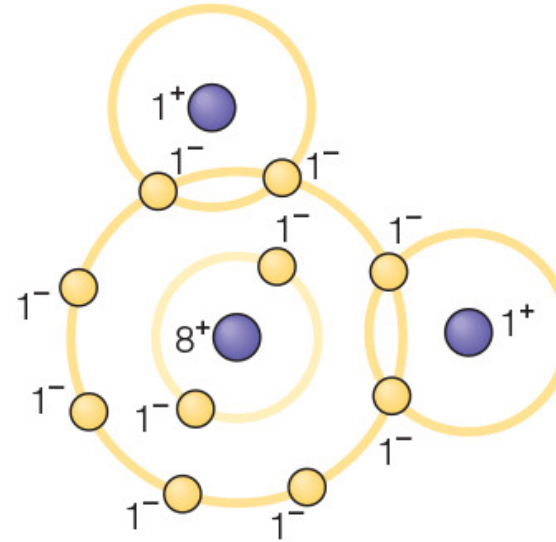
Electron
(1⁻ unit of charge)



share their electrons
with one oxygen atom . . .



to form a water molecule held
together by covalent bonds . . .



which acts as if it has
negative and positive ends.

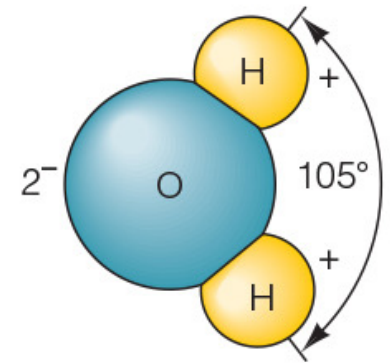
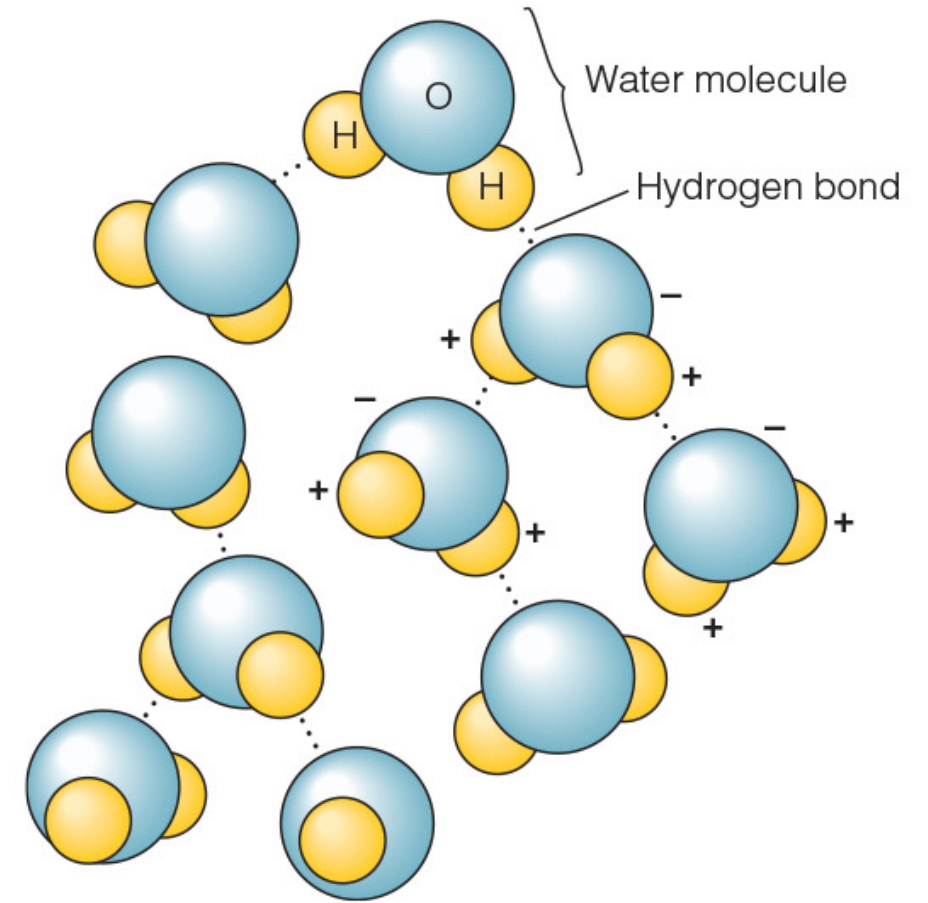


Figure 6.3 The formation of a water molecule. Atoms are most stable when they have a full outer shell of electrons. Hydrogen is most stable when it has two electrons in its outer shell and the larger oxygen atom (like many of the atoms relevant to our discussion of oceanography), is typically most stable with a total of 8 electrons in its outer shell. Since oxygen only has six electrons in its outer shell (because the first two of its eight electrons must fill the inner shell), oxygen bonds to two different hydrogen atoms by sharing hydrogen's electrons so that each atom has a full outer shell.

Due to its polarity water molecules are capable of forming Hydrogen Bonds

Hydrogen Bonds: They link water molecules together by electrostatic forces

Hydrogen Bonds are about 5-10% as strong as covalent bonds within the molecule, forming a webwork of molecules with very peculiar chemical and physical properties



- a** Hydrogen bonds in liquid water. The attractions between adjacent polar water molecules form a webwork of hydrogen bonds. These bonds are responsible for cohesion and adhesion, the properties of water that cause surface tension and wetting. Hydrogen bonds among water molecules also make it difficult for individual molecules to escape from the surface.

The polarity also allows the water to be a powerful solvent

The power of Hydrogen Bonds

- Cohesion – water molecules stick to each other, creating surface tension
- Adhesion – water molecules stick to other substances and make them wet

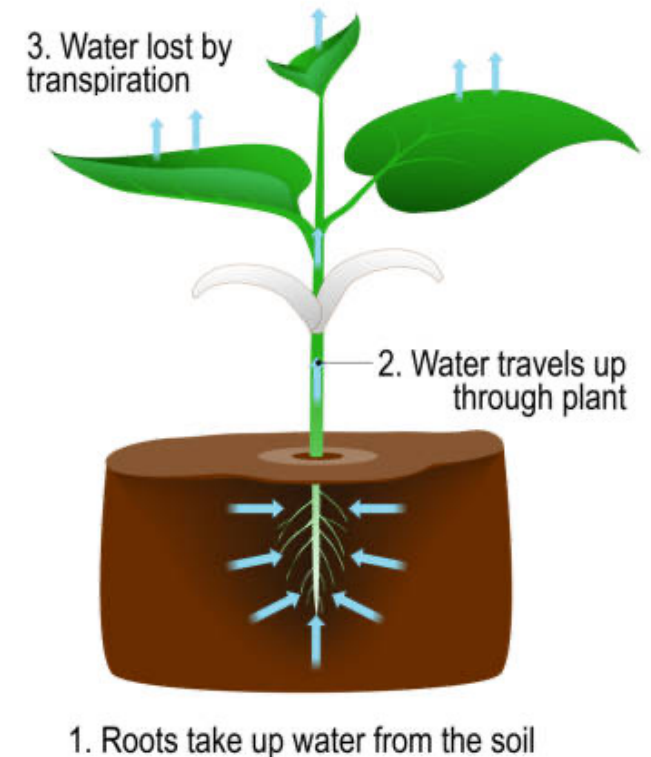
Causes of capillary action: From absorbent paper towels to plants hydration.



Prokuronov Andrey/Shutterstock.com

b Surface tension allows small bugs to be able to walk on water.

Figure 6.4 Hydrogen bonds and surface tension.



The power of Hydrogen Bonds

- Cohesion – water molecules stick to each other, creating surface tension
- Adhesion – water molecules stick to other substances and make them wet
- Molecules vibration push the neighbors absorbing more red light, allowing more blue light to get to us.

Iceberg



The power of Hydrogen Bonds

- Cohesion – water molecules stick to each other, creating surface tension
- Adhesion – water molecules stick to other substances and make them wet
- Molecules vibration push the neighbors absorbing more red light, allowing more blue light to get to us.
- **Water's state: Ice (solid), liquid water, water vapor (gas). Without the hydrogen bounds, liquid water would never be possible (gas or solid).**

The power of Hydrogen Bonds

- Cohesion – water molecules stick to each other, creating surface tension
- Adhesion – water molecules stick to other substances and make them wet
- Molecules vibration push the neighbors absorbing more red light, allowing more blue light to get to us.
- Water's state: Ice (solid), liquid water, water vapor (gas). Without the hydrogen bounds, liquid water would never be possible (gas or solid).
- **High Heat Capacity = It requires a lot of energy (heat) to change the water's temperature.**

This property is unusual and very important for the ocean circulation, climate, and life on Earth

An aerial photograph of a tropical coastline. The top half of the image shows deep blue ocean water. Below it is a narrow strip of white sand beach, followed by a dense line of green vegetation. The bottom half of the image shows a large area of light green, textured land, possibly a wetland or a large field. The text "Thermal Properties of Water" is overlaid in white, centered horizontally.

Thermal Properties of Water

Before we dive into the properties: What is the difference between Heat and Temperature?

Before we dive into the properties: What is the difference between Heat and Temperature?

- **Heat – energy produced by vibration of atoms/molecules**
- **Temperature – object's response to input or removal of heat (or vibration). Measure of how much energy there is in the system.**

Before we dive into the properties: What is the difference between Heat and Temperature?

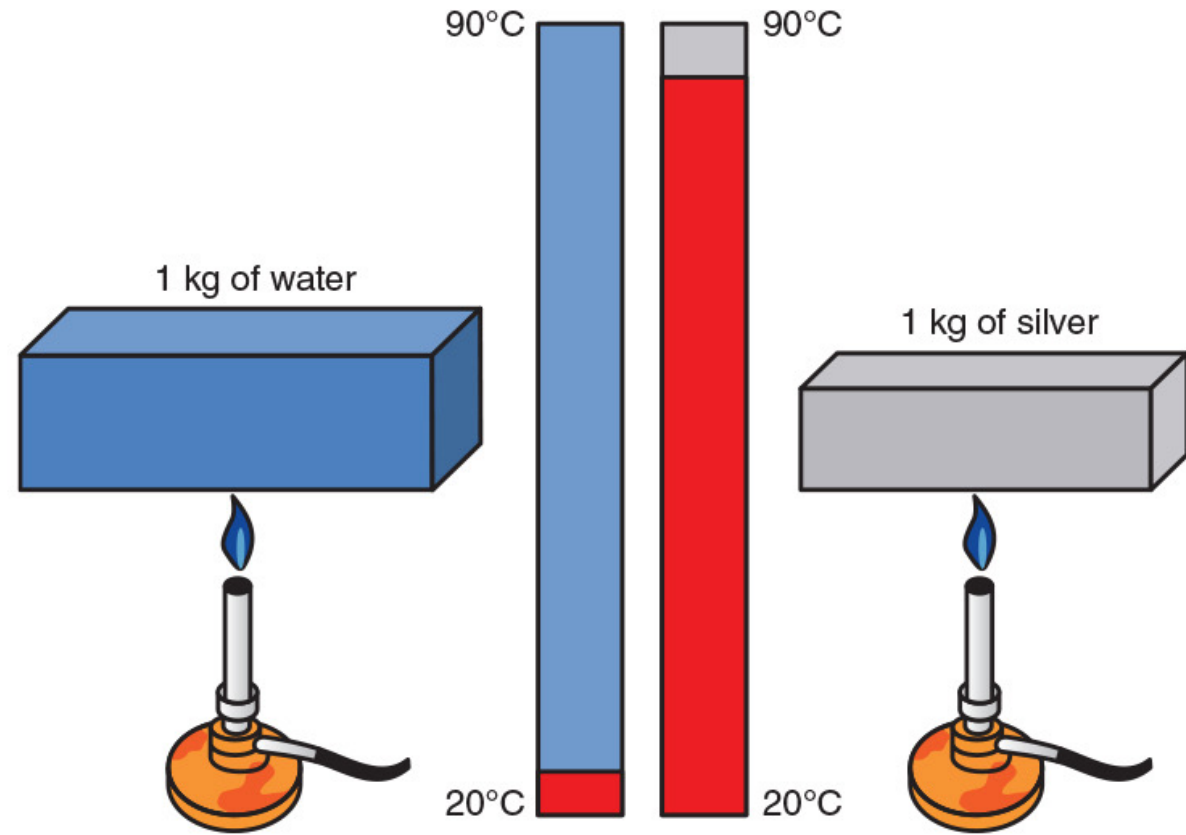
- **Heat – energy produced by vibration of atoms/molecules**
- **Temperature – object's response to input or removal of heat (or vibration). Measure of how much energy there is in the system.**
- **Heat capacity = Amount of energy necessary to change the temperature of a certain mass of a material.**
- **Water has high heat capacity (same as previous slide in other words) = Can absorb or release large amounts of heat with no small change in temperature**

Water Especial Thermal Characteristics

Substance	Specific Heat Capacity in calories/gram/°C
Silver	0.06
Granite	0.20
Aluminum	0.22
Alcohol (ethyl)	0.30
Gasoline	0.50
Acetone	0.51
Pure water	1.00
Ammonia (liquid)	1.13

- a** Heat capacity is a measure of the heat required to raise the temperature of a substance by 1°C (1.8°F). When comparing the heat capacity of different substances per unit mass, it is often referred to as the *specific heat capacity*, or *specific heat*, of a substance. Notice how little heat is required to raise the temperature of 1 gram of silver 1° in comparison with 1 gram of water.

Figure 6.5 Heat capacity of common substances.



- b** Temperature change in one kilogram of water and silver when exposed to the same amount of heat.

Quick Concept check:

1. Which material at 20° C has more internal energy (i.e., molecules vibration)?
- 1 kg of Silver
1 kg of Water

2. Using the concept of Heat Capacity (C_p), can you confirm your answer? Calculate the energy of both materials.

$$\text{Silver } C_p = 0.06 \frac{\text{calories}}{\text{gram} \cdot ^\circ\text{C}}$$

$$\text{Pure Water } C_p = 1.0 \frac{\text{calories}}{\text{gram} \cdot ^\circ\text{C}}$$

3. Are there any connections between the physical state of each material at 20° C (68° F) and the amount of vibration of the molecules?

Quick Concept check:

1. Which material at 20° C has more internal energy (i.e., molecules vibration)?
- 1 kg of Silver
1 kg of Water

Water

2. Using the concept of Heat Capacity (C_p), can you confirm your answer? Calculate the energy of both materials.

$$\text{Silver } C_p = 0.06 \frac{\text{calories}}{\text{gram} \cdot ^\circ\text{C}}$$

Silver energy = 1200 calories

$$\text{Pure Water } C_p = 1.0 \frac{\text{calories}}{\text{gram} \cdot ^\circ\text{C}}$$

Water energy = 20000 calories

3. Are there any connections between the physical state (solid, liquid, gas) of each material at 20° C (68° F) and the amount of vibration of the molecules?

At 20° C water is liquid and silver is solid. The rigid structure of solid materials is given by the locking of molecules into a strict geometrical order (almost no motion). In liquids and gases, the molecules are “free” to move around.

Latent vs Sensible Heat

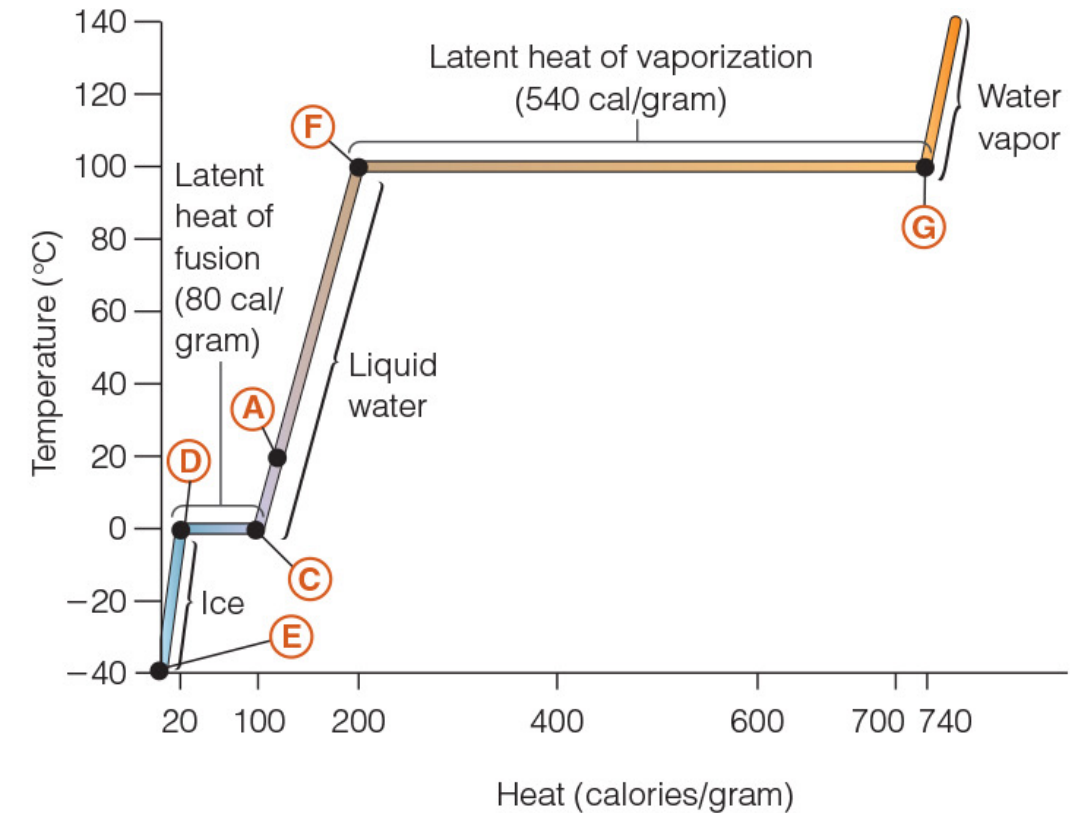


Figure 6.9 A graph of temperature versus heat as water freezes, melts, and vaporizes. The horizontal line between points **C** and **D** represents the latent heat of fusion, when heat is being added or removed but the temperature is not changing. The horizontal line between points **G** and **F** represents the latent heat of vaporization, when heat is being added or removed but the temperature is not changing. (Note that points **A–E** on this graph are the same as those in Figure 6.6.)

Latent vs Sensible Heat

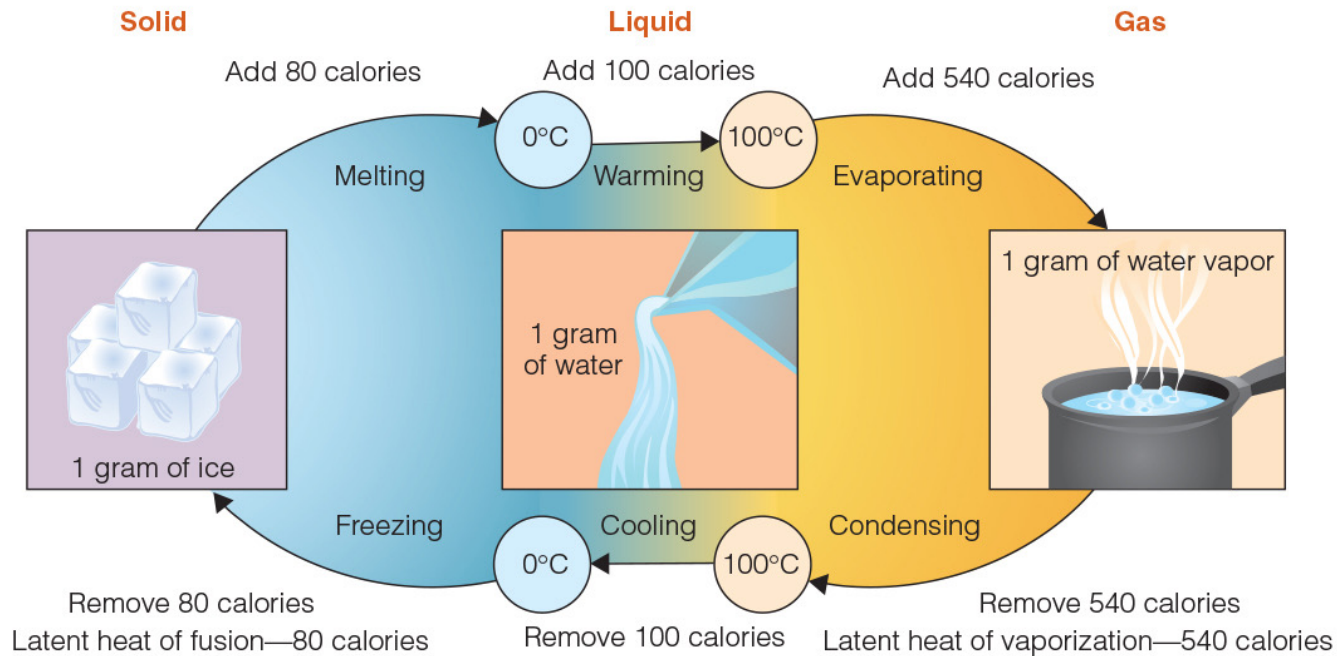


Figure 6.11 We must add 80 calories of heat energy to change a gram of ice to liquid water. After the ice is melted, about 1 calorie of heat is needed to raise each gram of water by 1°C. But 540 calories must be added to each gram of water to vaporize it—to boil it away. The process is reversed for condensation and freezing.

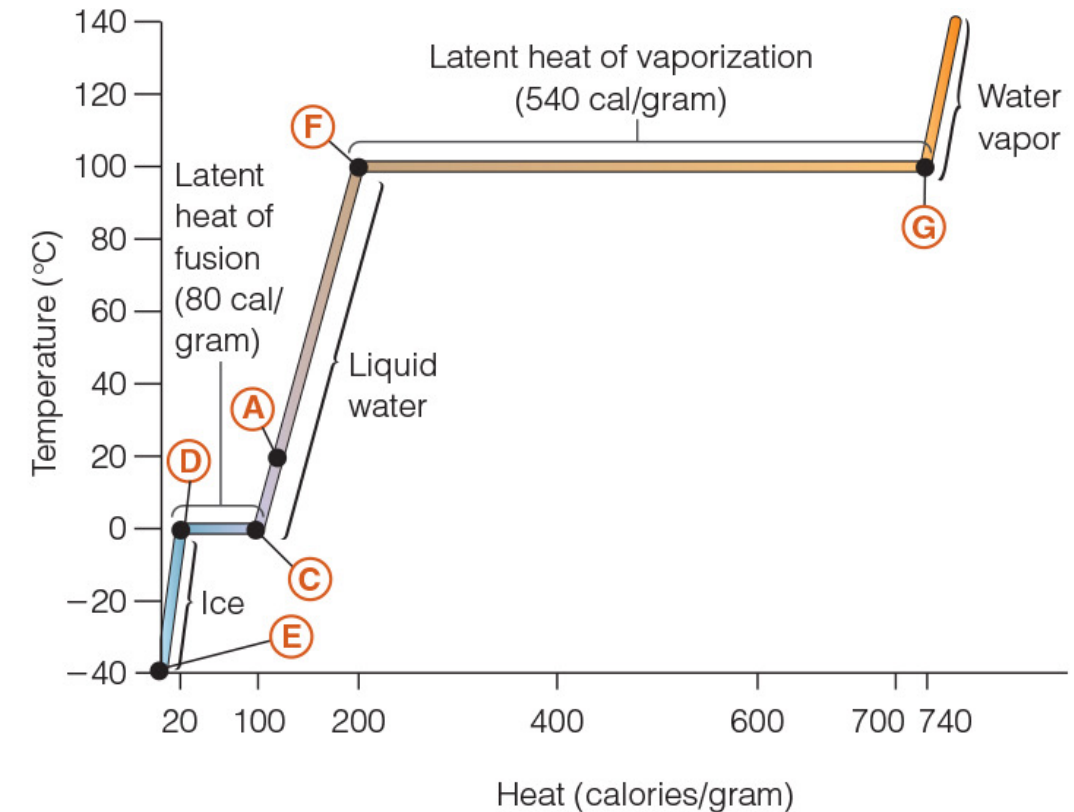
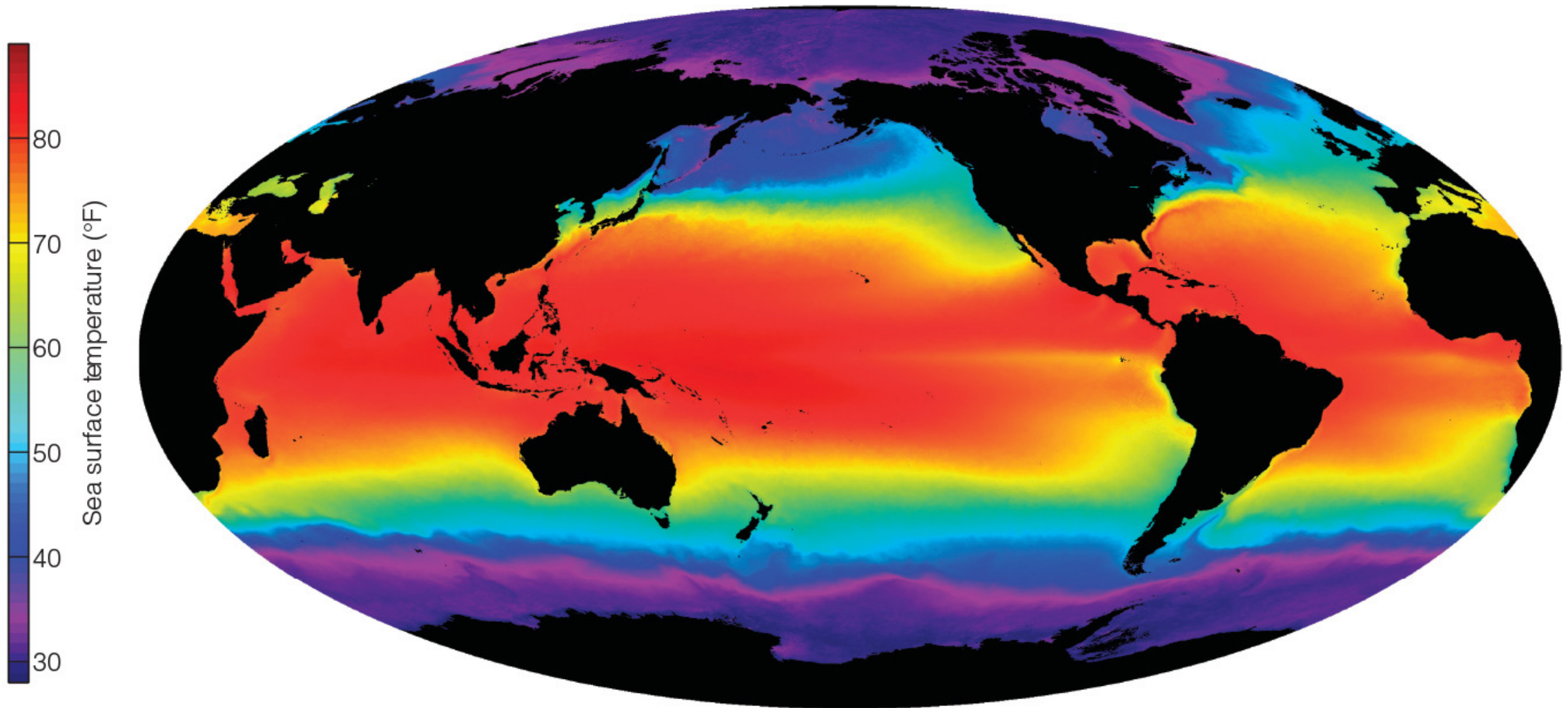


Figure 6.9 A graph of temperature versus heat as water freezes, melts, and vaporizes. The horizontal line between points **C** and **D** represents the latent heat of fusion, when heat is being added or removed but the temperature is not changing. The horizontal line between points **G** and **F** represents the latent heat of vaporization, when heat is being added or removed but the temperature is not changing. (Note that points **A**–**E** on this graph are the same as those in Figure 6.6.)

Why do we care about such properties?



Typical Sea Surface Temperatures observed using satellites

Ocean Moderates Global Climate (even over the continents, remember the icebreaker exercise!)

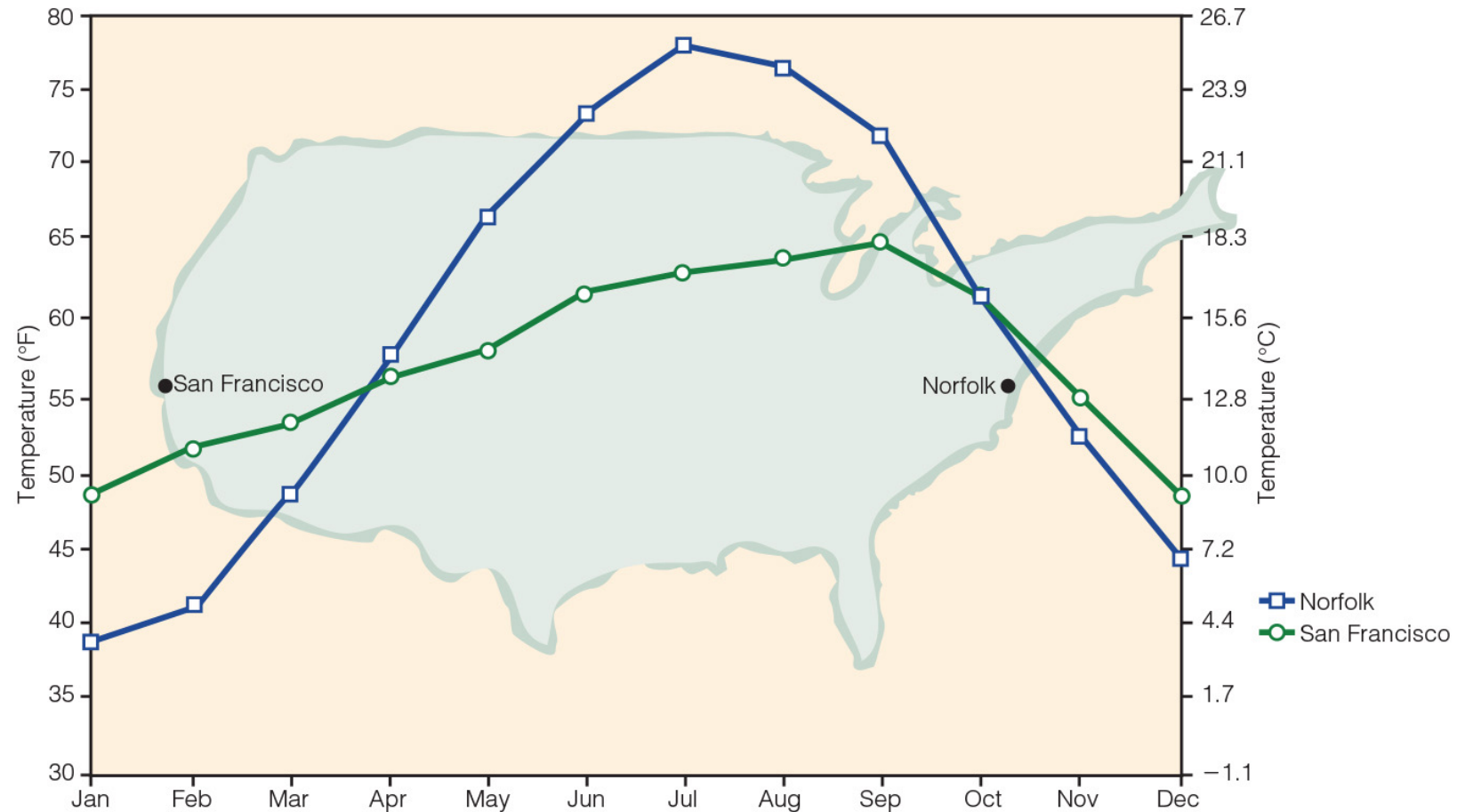


Figure 6.13 San Francisco, California, and Norfolk, Virginia, are on the same line of latitude, yet San Francisco is warmer in the winter and cooler in the summer than Norfolk. Part of the reason is that wind tends to flow from west to east at this latitude. Thus, air in San Francisco has moved over the ocean whereas air in Norfolk has approached over land. Water does not warm as much as land in the summer, nor cool as much in winter—a demonstration of thermal inertia.

Ocean Moderates Global Climate (even over the continents, remember the icebreaker exercise!)

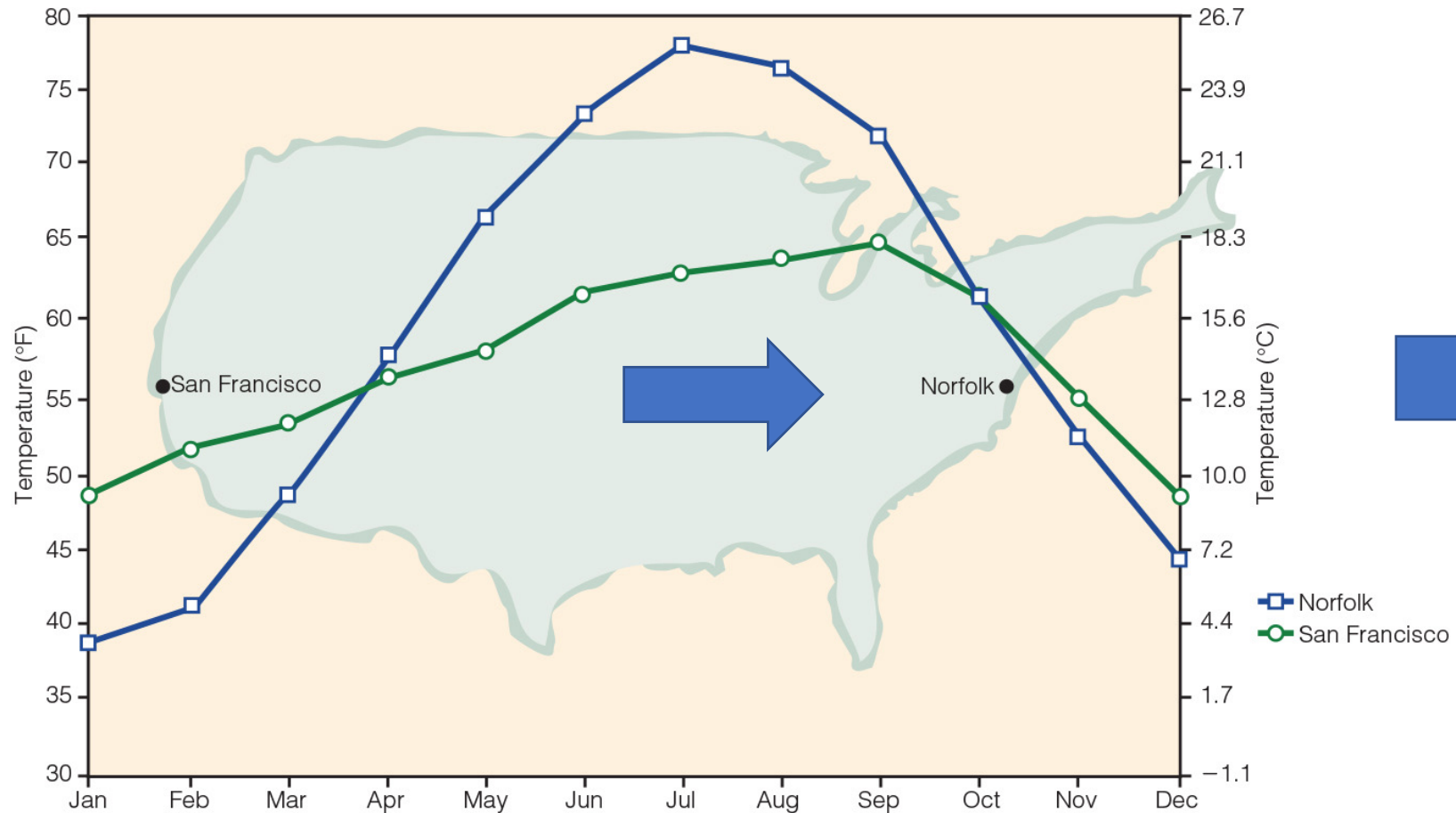


Figure 6.13 San Francisco, California, and Norfolk, Virginia, are on the same line of latitude, yet San Francisco is warmer in the winter and cooler in the summer than Norfolk. Part of the reason is that wind tends to flow from west to east at this latitude. Thus, air in San Francisco has moved over the ocean whereas air in Norfolk has approached over land. Water does not warm as much as land in the summer, nor cool as much in winter—a demonstration of thermal inertia.

An aerial photograph of a tropical coastline. The top half of the image shows deep blue ocean water. Below this is a narrow strip of white sand beach, followed by a shallow lagoon with turquoise water and visible sandbars. The bottom half of the image shows a dense green forest covering the land. The text is overlaid on the image.

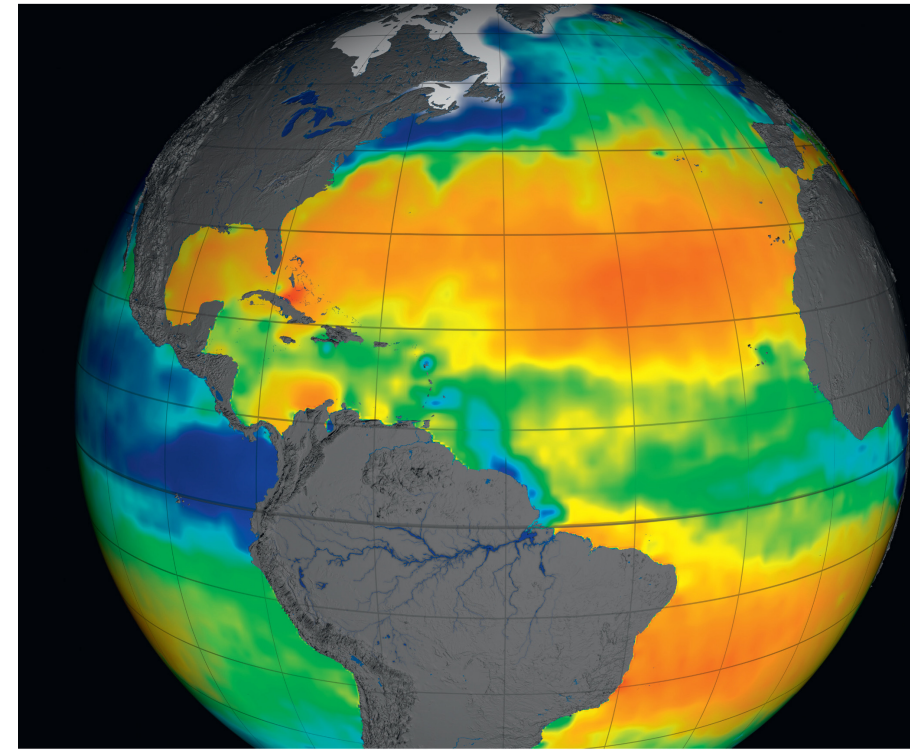
Salinity

As a result of the polar characteristic of the water molecule, the ocean can dissolve a lot of salt. Basically, water breaks the ionic bonds (connection where atoms exchange electrons making salts polar) that hold salt crystals together.

Salts originate in rocks and volcanic activity, accumulating in the ocean (similarly to sediments). However, in contrast to sediments, they are dissolved, and play a more important role in water properties and circulation.

Key Concept: Salinity = grams of salt dissolved in 1 kg of water (aka parts per thousand ‰)

Sea Surface Salinities from satellites measurements.
Red colors are higher salinities.



An aerial photograph of a tropical coastline. In the foreground, there is a wide, light-colored sandy beach. To the left of the beach, a shallow lagoon with clear, turquoise water is visible, bordered by a line of green vegetation. Further out, a coral reef is visible, with various shades of blue and green indicating different depths and reef structures. The water transitions from a light turquoise near the shore to a deep, dark blue further out. The overall scene is a beautiful representation of a healthy coastal ecosystem.

Density of Seawater

Density = mass/volume

Density mainly depends on Temperature (heat), but also on Salinity, and Pressure

Temperature impact on density: Thermal expansion and contraction of water

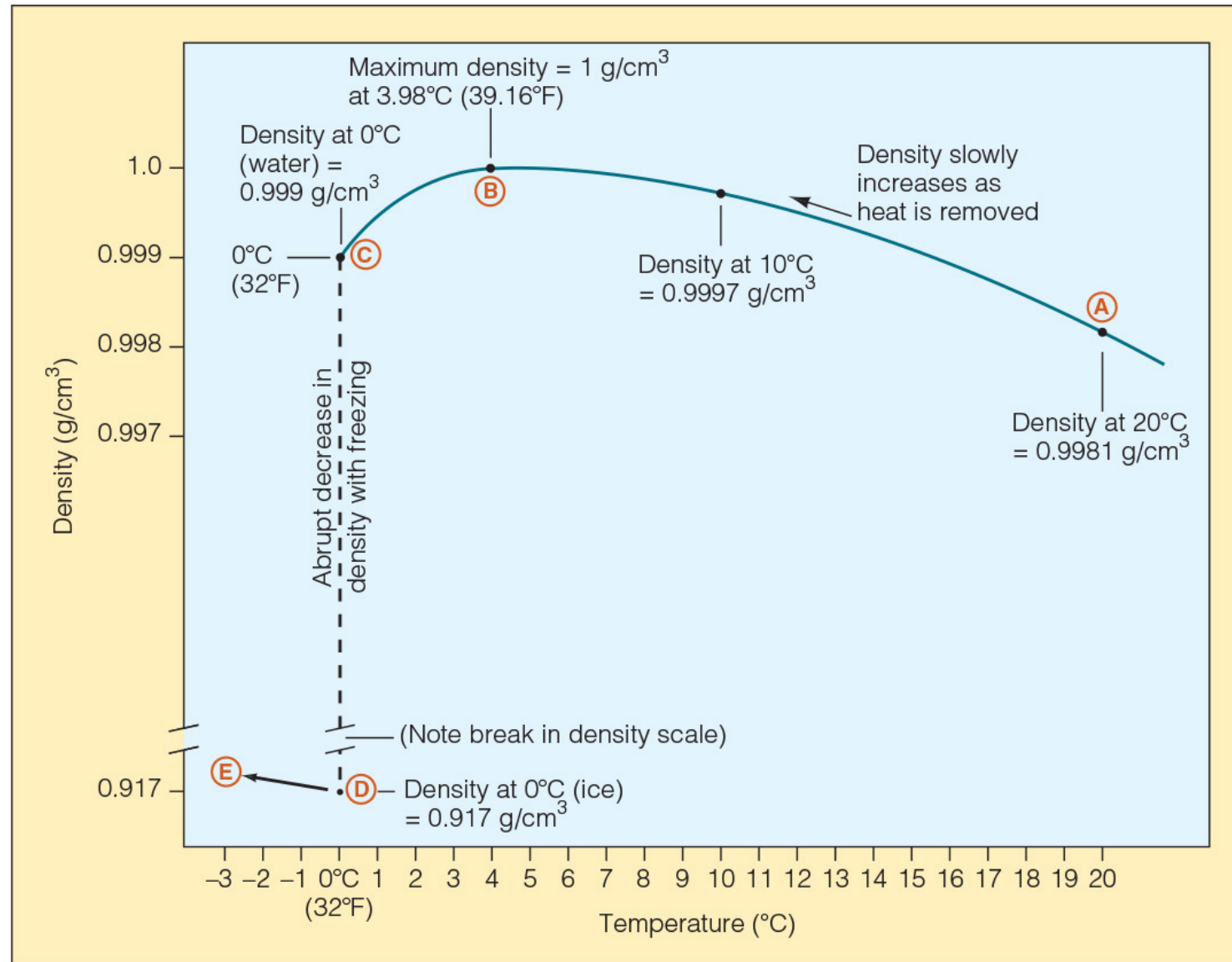


Figure 6.6 The relationship of density and temperature for pure water. Note that points **C** and **D** both represent 0°C (32°F) but show different densities, and thus different states of water. Ice floats because the density of ice is lower than the density of liquid water.

Temperature impact on density: Thermal expansion and contraction of water

Usually, molecules tend to occupy more space (volume) if they are vibrating more vigorously.

At temperatures lower than 4°C, the reorganization of the molecules make the water expand.

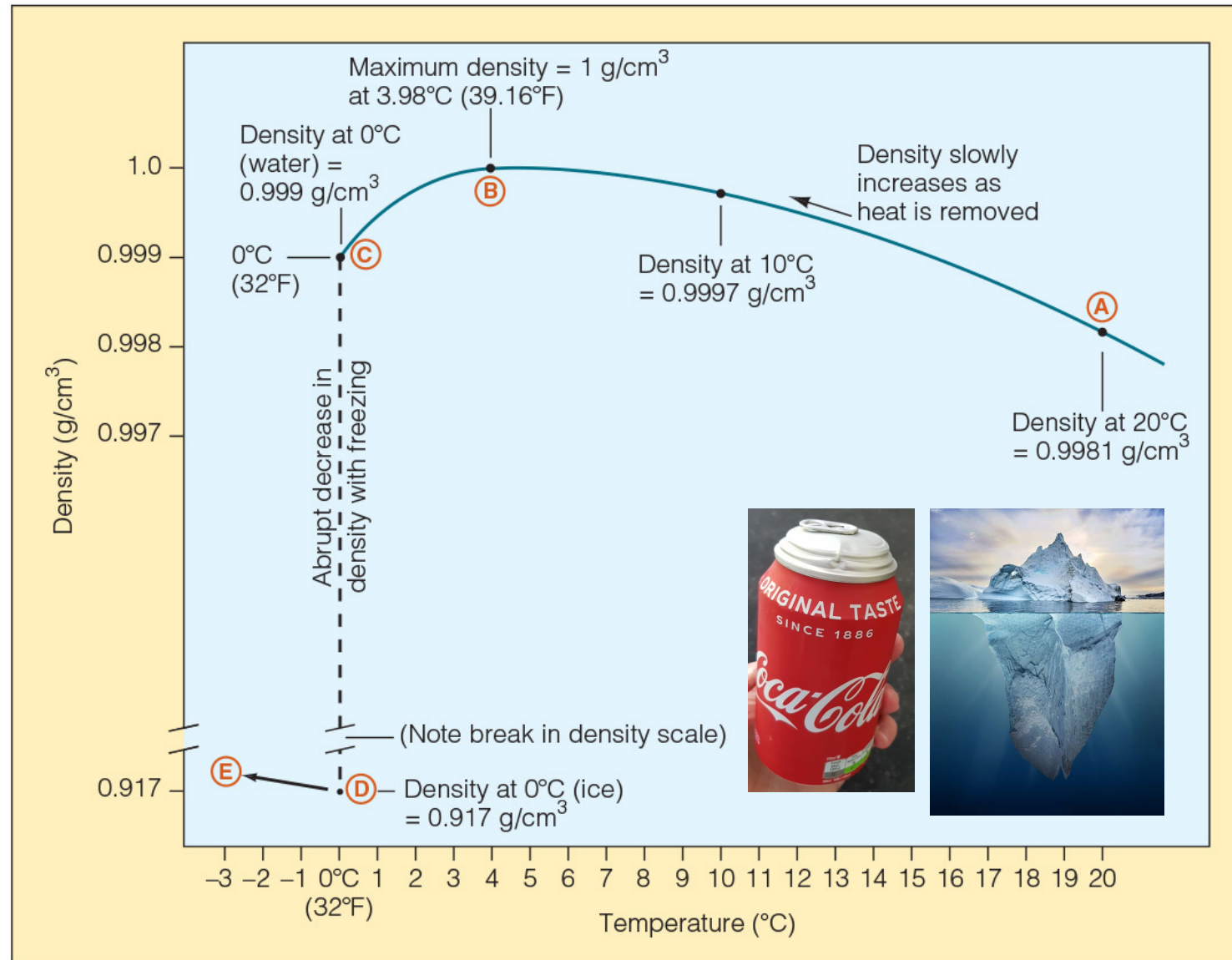


Figure 6.6 The relationship of density and temperature for pure water. Note that points **C** and **D** both represent 0°C (32°F) but show different densities, and thus different states of water. Ice floats because the density of ice is lower than the density of liquid water.

When we account for salinity: More salt, more mass, and higher density

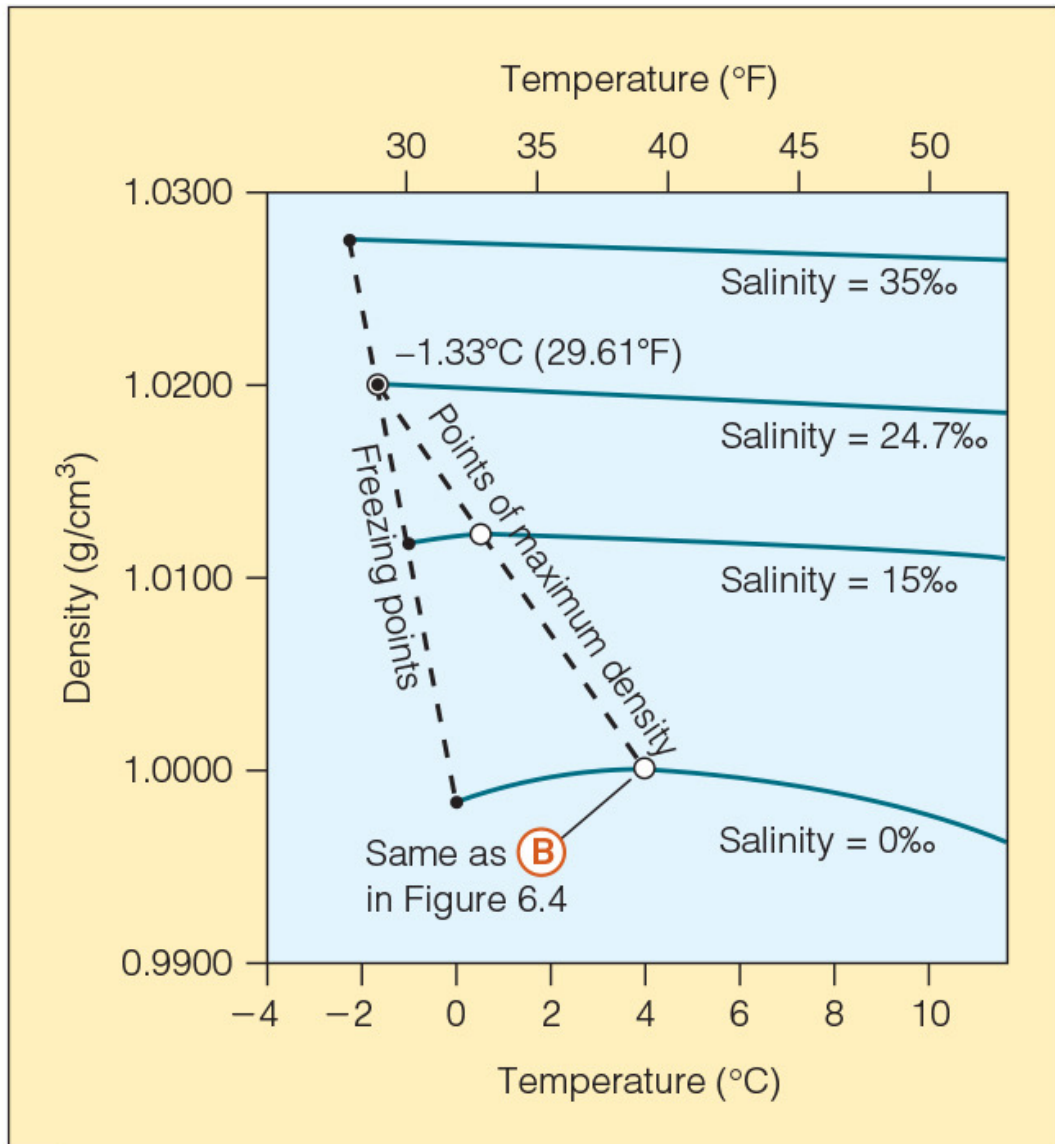
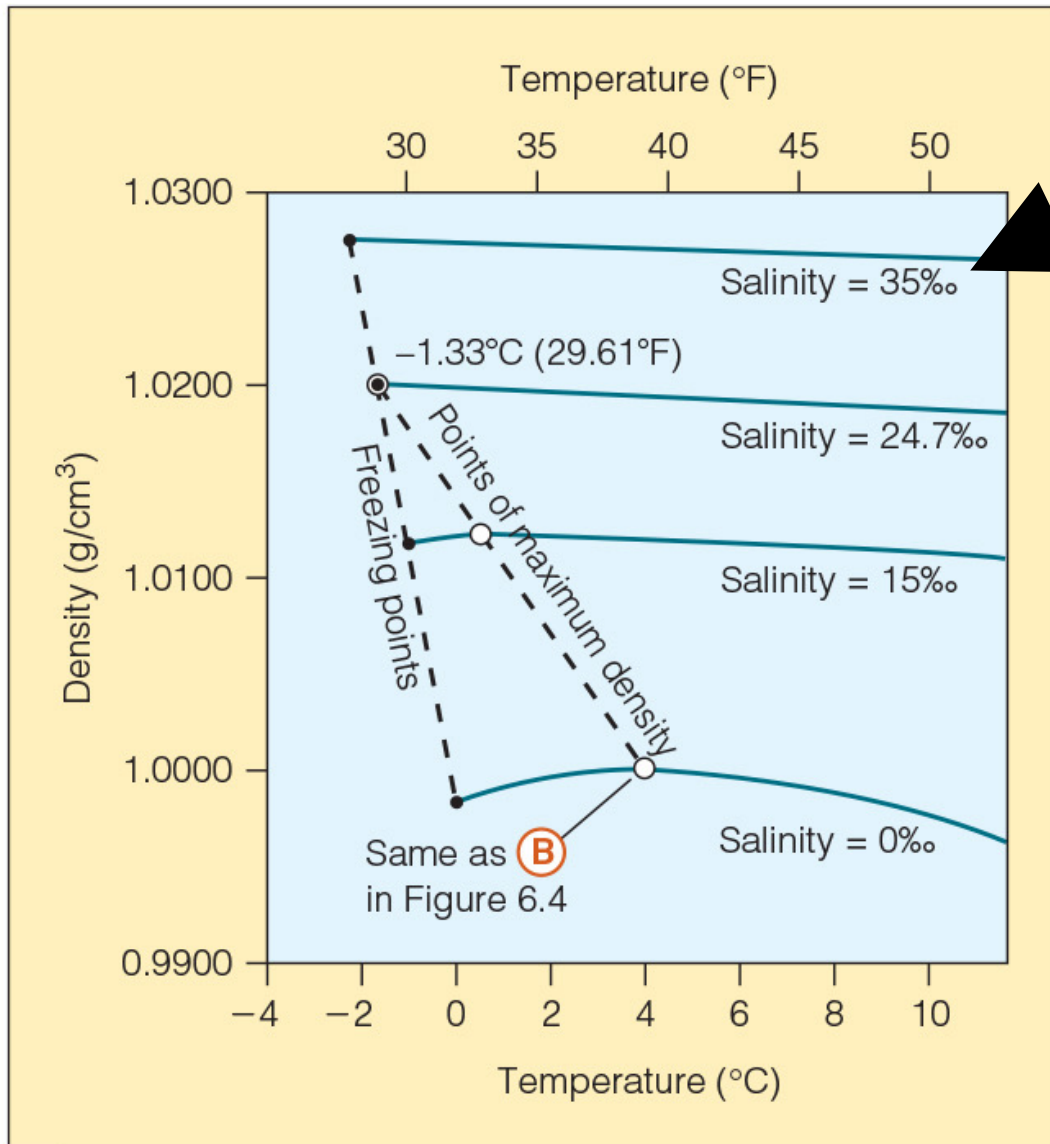


Figure 6.12 The dependence of freezing temperature and temperature of maximum density upon salinity (salt content). As we saw in Figure 6.6, pure water is densest at 3.98°C (39.2°F) (point B), and its freezing point is 0°C (32°F). Seawater with 15‰ salinity is densest at 0.73°C (33.31°F), and its freezing point is -0.80°C (30.56°F). The temperature of maximum density and the freezing point coincide at -1.33°C (29.61°F) in seawater with a salinity of 24.7‰. At salinities greater than 24.7‰ the density of water always decreases as temperature increases. Note that the symbol ‰ represents parts per thousand, so 2.47‰; 24.7‰.

When we account for salinity: More salt, more mass, and higher density



Typical ocean salinity

Salts make the water more “resistant” to freezing and evaporation, and the ions affect the restructuration of the molecules in low temperatures.

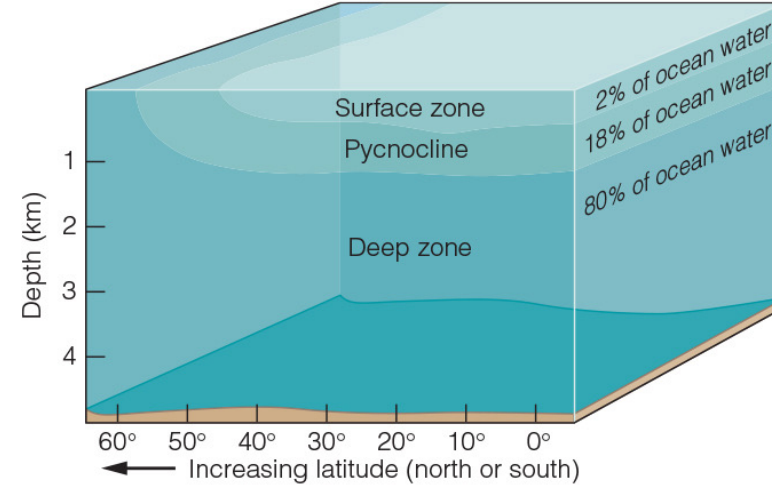
Pressure: It is worth mentioning that pressure pushes water molecules together, making it more compact.

Figure 6.12 The dependence of freezing temperature and temperature of maximum density upon salinity (salt content). As we saw in Figure 6.6, pure water is densest at 3.98°C (39.2°F) (point B), and its freezing point is 0°C (32°F). Seawater with 15‰ salinity is densest at 0.73°C (33.31°F), and its freezing point is -0.80°C (30.56°F). The temperature of maximum density and the freezing point coincide at -1.33°C (29.61°F) in seawater with a salinity of 24.7‰. At salinities greater than 24.7‰ the density of water always decreases as temperature increases. Note that the symbol ‰ represents parts per thousand, so 2.47%; 24.7‰.

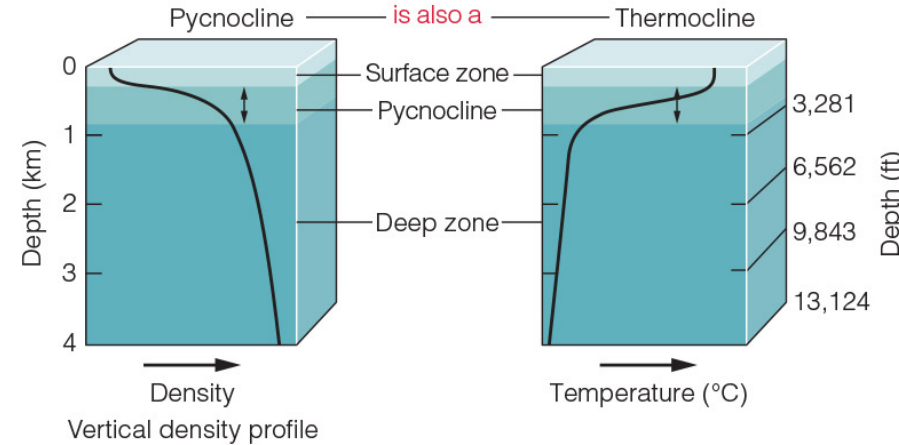
Ocean stratification: Yes layers!

Like the Earth, lighter layers fluctuate over denser layers.

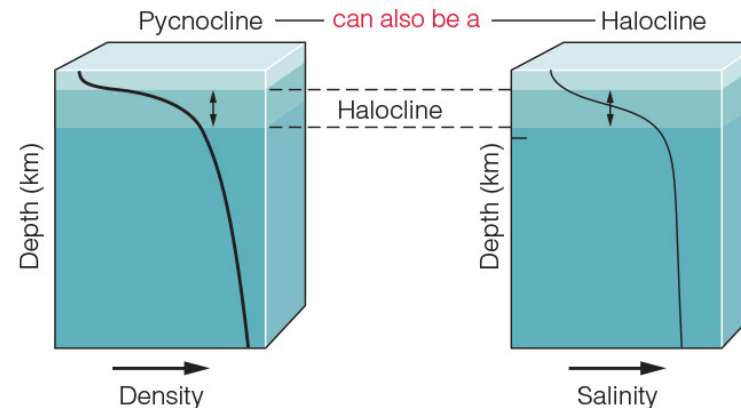
This layered (or stratified) structure constrain vertical motions in the ocean. Ocean currents tend to move more horizontally. It also impacts how light and sound move in the water (refraction).



- a** In most of the ocean, a surface zone (or mixed layer) of relatively warm, low-density water overlies a layer called the *pycnocline*. Density increases rapidly with depth in the pycnocline. Below the pycnocline lies the deep zone of cold, dense water—about 80% of total ocean volume.



- b** The rapid density increase in the pycnocline is mainly due to a decrease in temperature with depth in this area—the *thermocline*.



- c** In some regions, especially in shallow water near rivers, a pycnocline may develop in which the density increase with depth is due to vertical variations in salinity. In this case, the pycnocline is a *halocline*.

Figure 6.19 Density stratification in the ocean.

The power of ocean stratification from an easy home experiment

<https://www.youtube.com/watch?v=dSOv1KUVmtY>

Do the layers look the same everywhere?

Remember: Temperature usually dominates ocean density

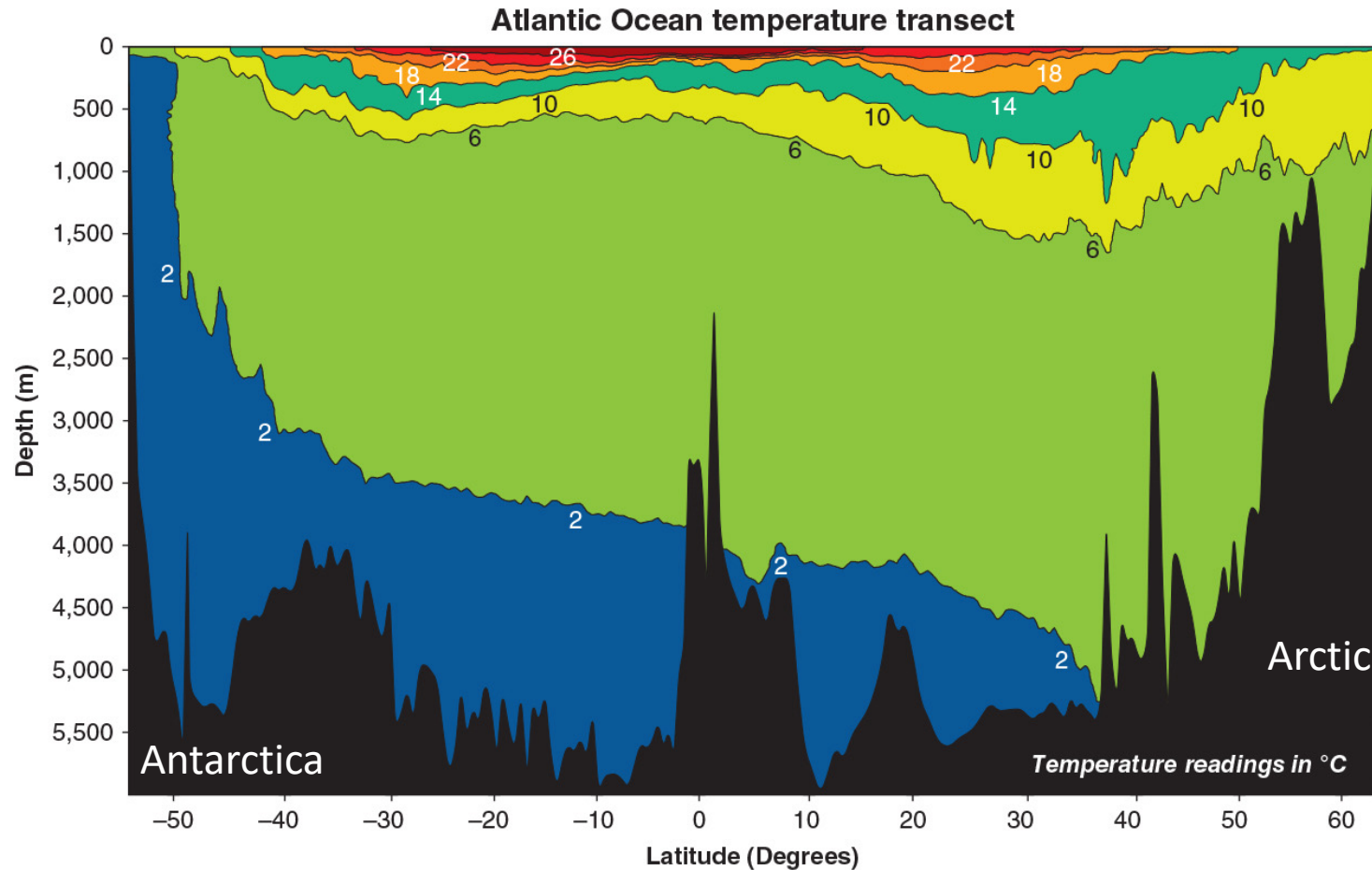


Figure 6.20 Temperature and salinity across the Atlantic Ocean basin at about 30° Degrees west longitude. Most seawater lies in cold darkness. In this image, a small and thin warm layer, concentrated near the equator, stands out above the relatively cool and consistently stable deep waters. As Earth warms, this layer will extend poleward. The irregular black shapes at the bottom of the graphs represent the ocean floor at the latitude shown.

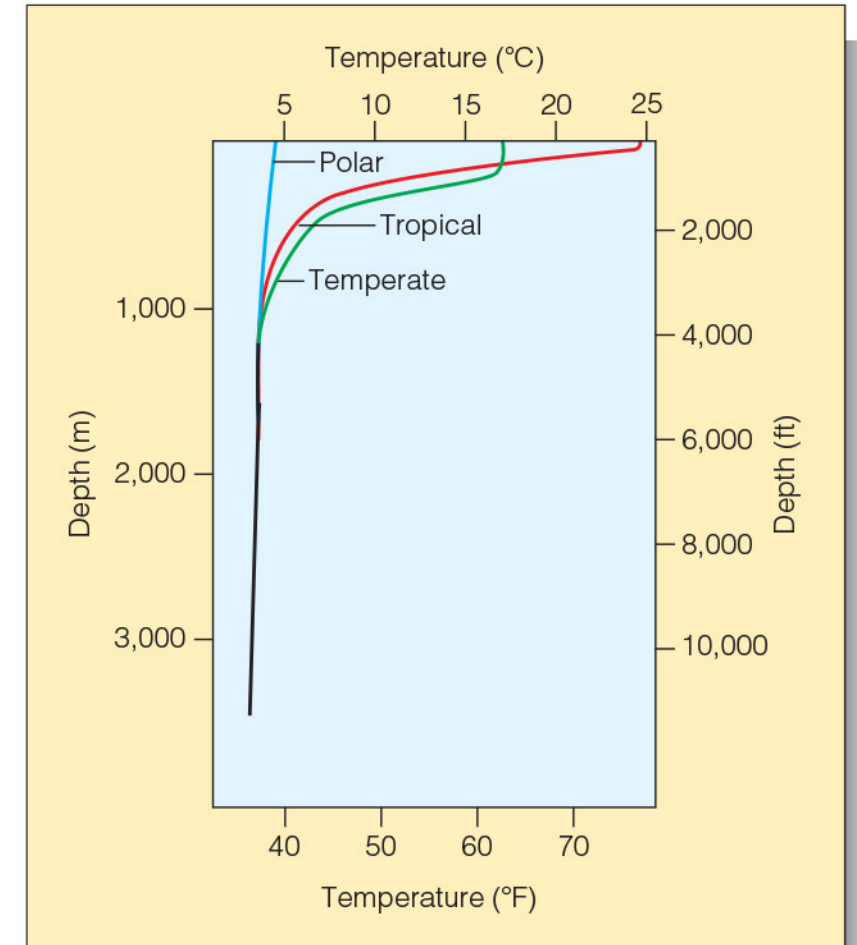


Figure 6.21 Typical temperature profiles at polar, tropical, and middle (temperate) latitudes. Note that polar waters lack a strong thermocline.